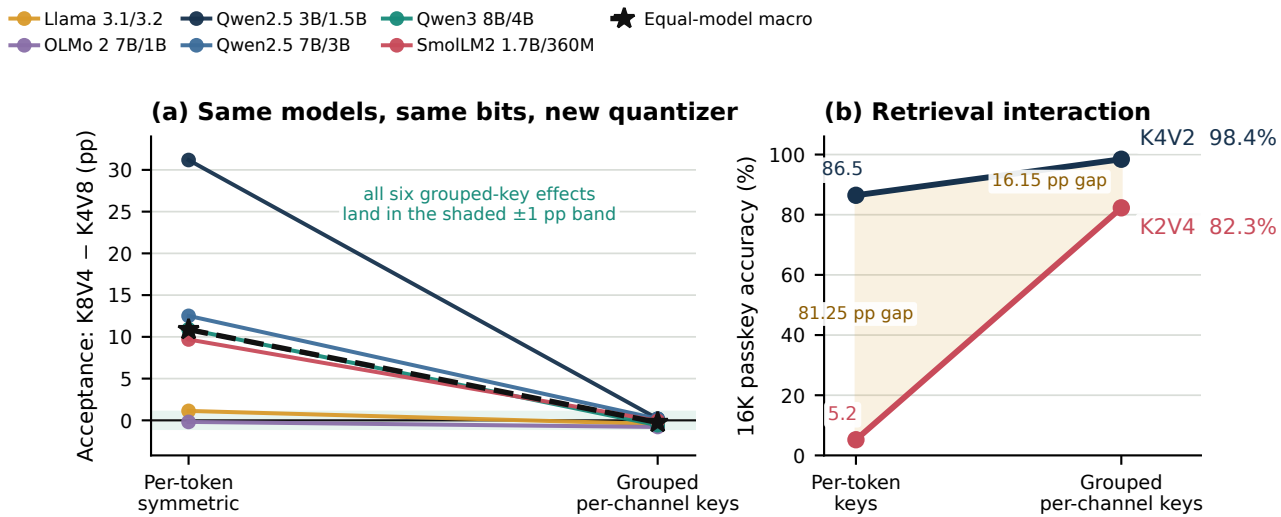
Llama 3.1/3.2
OLMo 2 7B/1B
Qwen2.5 3B/1.5B
Qwen2.5 7B/3B
Qwen3 8B/4B
SmolLM2 1.7B/360M
Equal-model macro
(a) Same models, same bits, new quantizer
Acceptance: K8V4 − K4V8 (pp)
all six grouped-key effects
land in the shaded ±1 pp band
Per-token
symmetric
Grouped
per-channel keys
(b) Retrieval interaction
16K passkey accuracy (%)
K4V2 98.4%
K2V4 82.3%
86.5
16.15 pp gap
81.25 pp gap
5.2
Per-token
keys
Grouped
per-channel keys